

① $d = 520$ $T = 625$ K $\delta = 600 \mu\text{m}$

a.) first find if $t > t^*$

$$t^*/d = 6.62 \times 10^{-7} e^{\left(\frac{11944}{T - 625}\right)} = 133 \text{ d}, \text{ so yes } t > t^*$$

then use δ^*

$$\delta^* = 5.1 e^{\left(\frac{-550}{625}\right)} = \boxed{2.115 \mu\text{m}}$$

$$\delta = \delta^* + K_L (t - t^*), \quad K_L = 7.48 e^{\left(\frac{-12500}{625}\right)} = 0.01542$$

$$= 2.115 + 0.01542 \left(\frac{520}{625} - 133 \right)$$

$$\boxed{\delta = 7.773 \mu\text{m}}$$

b.)

$$C_H^{cl} = \frac{2f \times \delta^* \rho_{\text{max}} \times \int_{\text{cm}^2}^{\circ} \times \frac{M_H}{M_0}}{\left(t - \frac{\delta}{\rho_B h} \right) \times \rho_{\text{max}}}$$

$$\delta_{\text{cm}} = 0.000773 \text{ cm}$$

$$t = 0.06 \text{ cm}$$

$$\delta = 0.000773 \text{ cm}$$

$$f_{\text{cm}}^{\circ} = \frac{32}{32 + 91} = 0.2601, \quad \frac{M_H}{M_0} = \frac{1}{16} = 0.0625$$

$$C = \frac{(2)(0.14)(0.000773 \text{ cm})(5.86 \text{ g/cm}^3)(0.2601)(0.0625)}{\left(0.06 - \frac{0.000773}{1.56} \right) (6.5 \text{ g/cm}^3)} = \frac{0.000202}{0.3868} \times 10^6$$

$$\boxed{C_H^{cl} = 53.26 \text{ wt. ppm}}$$

2.) Corrosion of Zr cladding is limited by oxygen diffusion through the ZrO_2 layer.

3.) Hydrides form ~~on the~~ towards the edge of the ~~core~~ zirconium cladding. They form during the oxidation process when oxygen interacts with the cladding. ~~Hydrogen~~ Hydrogen is pretty insoluble in Zr, so hydrides form. They also preferentially form closer to the edge due to lower temps / tensile stress. Hydrides lead to embrittlement, which diminishes the integrity of the cladding since cracks can occur easier.

4.) An RIA is a reactivity induced accident, most often introduced through control rod withdrawal. For PWRs, control rods can be ejected from the core, whereas for BWRs, control rods can fall out of the core. Limiting cases are both a zero-power, but for PWRs the case is at hot conditions at end of cycle. BWRs are limiting at STP and can be at any point in the cycle.

In an RIA, power spikes rapidly leading to ~~very high~~ a large temperature increase in the fuel. This causes rapid expansion and f.p. release, which can cause both fuel melt and/or cladding rupture.

- wanted some more

5.) A LOCA is a loss of coolant accident, whereby active cooling is lost. This leads to increase in cladding temperature that can lead to cladding rupture.

10/12

Between a LOCA and RIA, the RIA ~~leads~~ is a very short timescale event whereas, LOCA is on the order of minutes due to continuous heating from radioactive decay.

-wanted more

6.) Burnup in general increases failure probability during an accident. With ~~less~~ fuel at higher burnup, more corrosion has occurred leading to poorer heat transferability. Additionally, more hydriding has occurred leading to embrittlement. Radiation induced creep has further weakened the ^{more} cladding making it more susceptible to failure when exposed to accident conditions. Fission products inside the fuel matrix and plenum are at higher concentrations adding more strain on the fuel system.

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7.) 1. Improve cladding material properties

2. Improve fuel material properties

6/6 3. Improve fission product retention in the fuel

4. ~~limit~~ Reduce and slow down oxidation of the cladding

8.) One near term ATF-concept is doped fuel.

6/6 Adding small concentrations of metals such as Cr can improve material properties like thermal conductivity.

9.) Zirconium when introduced to high temperature steam can lead to oxidation. Oxidation is accelerated with increasing temperature and is an exothermic reaction. If the steam temperature is sufficiently high, a runaway reaction can occur.
5/6 - wanted more

10.) 1. Cladding oxidation ($< 17\%$ of original fuel thickness)

6/6 2. Power to melt (power to cause fuel melt, unlikely with oxide fuel)

3. Limit on LHR to prevent departure from nucleate boiling (DNBR)

11.) CRUD adds additional layers to the cladding surface that reduces heat transport out of the fuel.

5/4 Special to PWRs, CRUD can trap soluble boron which ~~shifts~~ ~~in into~~ induces power shifts (CIPS).
- radiation source term

12.) 1.) LiOH - Added to raise pH especially in PWRs to counteract soluble boron concentration (boric acid)

6/6 2. Ultrasonic cleaning - cleaning of fuel assemblies during refueling to remove corrosion / CRUD to limit CIPS and flaking off into primary system

7/8 13.) 1.) Restructuring - in max fuel, the structure of the fuel changes radically with increasing burnup. Assuming solid fuel, a central void region develops and grain size increases as you move inward

2) O/M ratio - in max fuel, the fuel is typically always manufactured to always be hypostoichiometric ($\text{o/m} < 2.0$) because the o/m increases as burnup increases.
- happens w/ UO_2 also...

